AskNow: An LLM-powered Interactive System for Real-Time Question Answering in Large-Scale Classrooms

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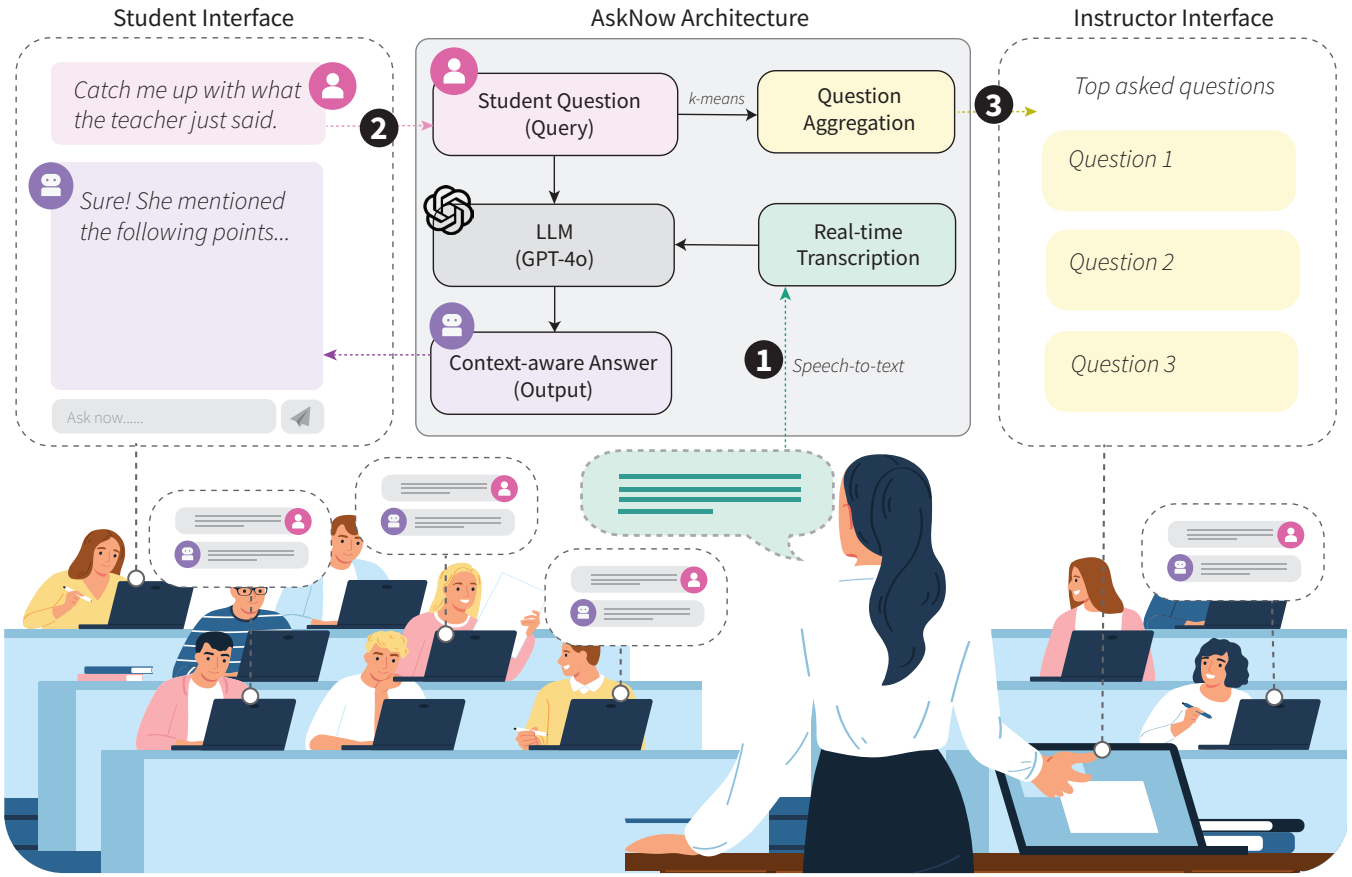


Figure 1: Overview of AskNow, an interactive system powered by LLM, consisting of Student Interface and Instructor Interface. In a large-scale classroom, 1) AskNow is listening to the instructor using streaming speech-to-text technology. 2) Students can ask questions through their own devices and get immediate feedback grounded in the ongoing lecture context. 3) Instructor Interface displays common confusion areas in real time.

Abstract

In large-scale classrooms, students often struggle to ask questions due to limited instructor attention and social pressure. Based on findings from a formative study with 24 students and 12 instructors, we designed AskNow, an LLM-powered system that enables students to ask questions and receive real-time, context-aware responses grounded in the ongoing lecture and that allows instructors to view students' questions collectively. We deployed AskNow in three university computer science courses for a week and tested with 117 students. To evaluate AskNow's responses, each instructor rated the perceived correctness and satisfaction of 100 randomly sampled AskNow-generated responses. In addition, we conducted interviews with 24 students and the three instructors to understand their experience with AskNow. We found that AskNow significantly reduced students' perceived time to resolve confusion. Instructors rated AskNow's responses as highly accurate and satisfactory. Instructor and student feedback provided insights into the role of such systems in supporting real-time learning in large lecture settings.

CCS Concepts

• **Human-centered computing → Interactive systems and tools**; • **Applied computing → Interactive learning environments**.

Keywords

Large Language Models (LLMs); Educational Technology; LLMs in Education; Real-time Question-Answering; Classroom Learning; Large-scale Classroom

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1 Introduction

Large-scale classrooms present unique challenges for student-instructor interactions, including students feeling reluctant to ask questions due to social pressure [10] or unable to attract instructor attention [27], challenges exacerbated by more advanced or fast-paced classes [9]. Instructors, on the other hand, can be overwhelmed with teaching while managing a large classroom and monitoring student attention, confusion, and questions [51]. Researchers and educators have explored various technology-mediated interventions to address these challenges. Audience Response Systems (ARS), such as Poll Everywhere, Slido, and Top Hat, allow students to participate anonymously through multiple-choice polling, enhancing engagement, attendance, and ability to give and receive immediate feedback [15, 38, 46]. Similarly, classroom backchannel systems, such as Hotseat [1] and Backchan.nl [33], enable students to post open-ended questions during lectures without interrupting the instructor [5, 49]. Although these tools effectively promote student participation and relay student confusion to instructors, these technologies still offer limited support to students, due to lack of personalized, real-time feedback, and instructors, due to the added demand on their attention [29].

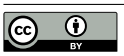
Recent advances in large language models (LLMs) have expanded possibilities for educational support across domains, including programming education [36, 42, 68, 75, 76], early childhood learning [35, 77], and tutoring [28]. Systems, such as SPHERE, assist instructors in reviewing and prioritizing feedback on student code [68], while a virtual teaching assistant, named Jill Watson, answers questions in online education [67]. Although these systems demonstrate the potential of LLMs for personalized feedback and support, they are primarily designed for asynchronous interactions or specialized contexts such as real-time code assessment. Little work has been done to explore how LLMs could provide students and instructors with contextualized, real-time support during synchronous, in-person lectures, particularly in large-scale classroom settings.

To address these challenges, we conducted a two-phase study. In Phase 1, we interviewed 24 students and 12 instructors to understand their experiences with learning and teaching in large classrooms, respectively. In particular, we discussed challenges in real-time interaction, strategies for resolving confusion, and expectations for AI-powered support tools. To address the challenges identified in our formative study, we designed AskNow, an interactive system that leverages LLMs to provide students with immediate, context-aware answers to their questions during live lectures. AskNow synchronously records and transcribes the lecture content, which serves as context for the LLM's responses. In Phase 2, we deployed AskNow across three large university computer science courses of varying difficulty levels, where 117 students used AskNow over two sessions of their class and responded to pre- and post-surveys measuring perceived effectiveness of AskNow. We further evaluated the system through expert assessments, where instructors rated the accuracy of AskNow-generated responses. Finally, we interviewed all three instructors and 24 students regarding their experience using AskNow.

Our data showed that AskNow significantly reduced students' perceived time to resolve confusion during lectures. Instructors rated AskNow's responses as highly accurate and satisfactory. Students demonstrated a variety of use patterns based on when and why they used AskNow to seek clarification, and instructor feedback revealed insights into how real-time AI systems could complement traditional classroom teaching to better support large-scale learning environments.

Our work makes the following contributions:

- *Formative Data*: An empirical understanding of large classroom needs from both instructor and student perspectives.
- *System Contribution*: AskNow, a real-time AI-based question-answering tool for large-scale classrooms.
- *Evaluation Data*: An empirical understanding of how AskNow was used across three university courses.
- *Design Implications*: Insights from our real-world deployment synthesized into implications to inform the design of future LLM-powered classroom support systems.



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2 Related Work

2.1 Challenges in Large-scale Classrooms

Although large-enrollment courses are often valued for their efficiency, they consistently constrain meaningful interaction and make it difficult for instructors to monitor individual student understanding [27, 30, 50]. Within crowded lecture halls, many students refrain from asking questions due to communication apprehension, which substantially limits overall participation [10, 51]. The time pressure of covering extensive material, coupled with the traditional single-speaker paradigm, further restricts opportunities for turn-taking, such that only a small subset of vocal students tends to dominate classroom discourse [27, 70]. As class size increases, empirical work has shown declines in active engagement and reductions in teacher–student interactions, with particularly pronounced effects for students with lower prior attainment [8]. Foundational analyses of lecturing similarly highlight that conventional lectures are primarily designed for efficient information transmission rather than adaptive, dialogic exchange [9, 31].

The pedagogical costs of these interaction gaps are considerable. Misconceptions often remain unvoiced during the rapid pace of delivery, and opportunities for in-the-moment clarification are lost [30, 50]. A substantial body of research underscores the importance of immediate feedback, demonstrating that students benefit when misunderstandings are addressed promptly rather than after delays [4, 23, 39, 65]. Interactive and team-based learning formats similarly rely on rapid clarification cycles, which have been linked to greater engagement and deeper knowledge construction [12, 59]. In sum, scale itself introduces structural barriers to inclusive participation and timely feedback, leaving widespread confusion easily overlooked in the context of large-scale lectures [2, 20, 27, 51, 65, 70].

2.2 Technology for Large-scale Classrooms

To mitigate challenges in large-scale classrooms, researchers and educators have explored various technology-mediated interventions. One prominent line of work is Audience Response Systems (ARS)–commonly known as clickers or live polling tools–such as Poll Everywhere, Top Hat, and traditional handheld clickers (*e.g.,* TurningPoint). These tools enable simultaneous student responses to multiple-choice questions, providing aggregated and real-time feedback to the class [15, 38, 46]. Many studies have demonstrated the positive impact of ARS in large-scale classrooms, including improved attendance via increased student motivation [13, 15, 32]; enhanced attention through interactive, real-time polling [7, 13, 15, 21]; and increased participation by leveraging anonymity to reduce anxieties associated with public speaking [61, 62, 66]. Furthermore, ARS usage correlates positively with overall classroom engagement, as students actively reflect and respond to lecture content rather than passively receiving information [22, 57]. Finally, empirical evidence consistently demonstrates that ARS can promote students' learning performance, knowledge retention, and academic achievement across diverse disciplines, including physics [11], biology [57], pharmacy education [63], medicine [56, 60], and general undergraduate instruction [37, 40]. These improvements are largely attributed to ARS's ability to facilitate frequent formative assessment, promote active engagement, and provide immediate feedback.

Another technology-based strategy is the classroom backchannel, a secondary communication stream enabling students to post questions, comments, or feedback in real-time alongside the primary lecture[1, 33]. Examples such as Hotseat at Purdue University [1], Backchan.nl [33], and Backstage [55] enable participation in large lectures via web or mobile interfaces. Hotseat, in particular, has shown effectiveness in lowering barriers related to anxiety or social pressure by offering anonymous participation [1]. Evaluations of backchannel systems indicated increased student interaction, especially during fast-paced or complex lectures, as students were able to ask questions without interrupting the instructor [5, 49]. Students also reported greater engagement due to the immediate visibility of their questions or feedback [1, 34]. Overall, backchannel tools offer students ways to resolve confusion and enhance their engagement during lectures.

However, despite their demonstrated benefits, both ARS and backchannel systems have notable limitations. ARS tools usually only allow students to respond through simple multiple-choice or polling interactions, limiting their ability to capture more detailed or complex student confusions. Backchannel systems, although supporting open-ended queries, still rely heavily on manual responses from instructors or moderators, hindering real-time interaction in large lectures. Furthermore, continuous monitoring of student inputs may increase instructor cognitive load, disrupting lecture flow [29]. In addition, students may get distracted from the ongoing lecture while posting questions through backchannel [1, 5]. Most importantly, neither type of system provides immediate, personalized, and detailed feedback, often leaving student misunderstandings unresolved during class.

2.3 AI-based Systems for Learning Support

Recent advancements in artificial intelligence (AI) and large language models (LLMs) have created new opportunities for enhancing education. With capabilities such as real-time natural language interaction, personalized feedback generation, and flexible content creation, these technologies offer the potential to support learning at scale and to augment traditional instruction with more adaptive, individualized support. Researchers have begun exploring how AI-driven systems can address challenges across various educational domains, improving both student learning experiences and teacher workflows.

In programming education, several systems have leveraged LLMs to support scalable learning. VizProg visualizes student coding progress to help identify misunderstandings [75], while CFlow extends this work by enabling large-scale semantic flow analysis of student code [76]. SPHERE builds on these approaches by reviewing LLM outputs to prioritize critical feedback for instructors [68]. Other applications have explored the use of LLMs for learning support beyond computing. Storyfier supports vocabulary acquisition through AI-generated storytelling [54], while LessonPlanner [24] and LitLinker [25] assist novice teachers in designing pedagogy-aligned lesson plans and interdisciplinary learning contexts, respectively. In early childhood education, Mathemyths introduces co-creative storytelling to teach mathematical language [77], SET-PAiREd enables parents and LLMs to co-design educational content

for robot-mediated learning [35], and StoryMate personalizes interactive story reading experiences for children [16]. For assessment and tutoring, systems such as ConQuer facilitate concept-based quiz generation [28], while Jill Watson acts as a virtual teaching assistant that answers student questions in online education based on course materials [67]. Building on this line of work, Tiwari et al. introduced an AI-assisted teaching support tool that automatically answers frequently asked questions [69], and Askademia uses a vision–language model to provide responses to student queries during lectures based on course materials and slides [48].

While these systems demonstrate the versatility of AI and LLMs in educational contexts, much of the current work has focused on asynchronous, one-on-one, or home-based learning environments. A small but growing body of work explores the integration of AI into real-time classroom settings. For example, Askademia and PairUp both investigate how real-time agents can support students and teachers during live sessions [48, 73]. However, these systems primarily evaluate correctness, efficiency, and orchestration, and provide limited instructor-facing awareness of students' collective confusion, offering little discussion of how such agents may influence students' in-class attention or perceptions of instructional authority. Subsequent work on AI co-orchestration tools further examines how teachers conceptualize sharing control with AI systems, highlighting concerns around trust, responsibility, and efficiency [41].

Despite these advances, there remains a significant gap in research on how LLMs might directly support students' real-time clarification needs during live lectures while also preserving instructor oversight and authority. Existing AI-based classroom systems primarily aim to assist instructors in managing class dynamics or provide delayed feedback, and even real-time virtual teaching assistants rarely give instructors an aggregated view of ongoing student confusion in large lecture halls. To address this gap, the present work explores the use of LLMs as real-time classroom assistants, supporting dynamic clarification in large-scale lectures.

3 Formative Study

Based on the prior work, we identified that instructors face challenges in gauging individual student understanding and facilitating real-time interaction, while students have difficulties in voicing questions and receiving immediate feedback during lectures. To inform how we can design a system to bridge these gaps, we conducted a formative study with 24 students and 12 instructors to understand their experiences in classroom learning, including the challenges they face, current strategies for resolving confusion, and their expectations for AI-powered support tools.

3.1 Methods

3.1.1 Participants. We recruited a total of 36 participants, including 24 students (13 female, 11 male; Age between 18 and 29, $Mean = 22.38$, $SD = 3.14$) and 12 instructors (4 female, 8 male), from a large public university in the Midwestern United States. Recruitment was conducted university-wide. Student participants were enrolled in a variety of disciplines and included both undergraduate and graduate students across all years of study. Instructor participants were similarly diverse, from different academic departments and teaching at least one course during the current academic semester. Demographic details of participants are presented in Table 1 (Students) and Table 2 (Instructors). Each Participant received $10 as compensation for completing the interview. The study was approved by our university's IRB.

3.1.2 Semi-structured Interviews. Each participating student and instructor took part in a semi-structured interview, which lasted for around 30 minutes. For both students and instructors, the interviews began with open-ended questions about recent classroom experiences, challenges, and current strategies for resolving confusion. Only in the latter part of the interview did we introduce a short vignette about a hypothetical AI assistant, using it to probe participants' expectations and concerns rather than to elicit concrete feature requests. Consistent with prior work highlighting that users may struggle to articulate their underlying needs directly [53], we anchored our questions in concrete past experiences and probed for the reasons behind participants' suggestions. Student interview questions explored their in-class confusion experiences (*e.g.,* "*Have you ever encountered* moments during the lecture when you don't

Table 1: Students' Demographics in the formative study

ID	Gender	Level of Study	Year of Study	Major(s)	Native English Speaker
S1	M	Undergrad	1	English Literature, Japanese	No
S2	M	Grad	1	Special Education	No
S3	F	Undergrad	1	Neurobiology, Psychology	No
S4	F	Grad	1	Biotechnology	No
S5	F	Grad	2	Endocrinology, Reproductive Physiology	No
S6	F	Undergrad	3	Chemical Engineering	Yes
S7	M	Grad	1	Business Administration	Yes
S8	M	Grad	2	Educational Policy Studies	No
S9	M	Grad	1	Information Science	Yes
S10	M	Undergrad	2	Biochemistry, Asian Languages and Cultures	Yes
S11	M	Undergrad	1	Journalism	Yes
S12	F	Grad	3	Industrial Engineering	No
S13	M	Undergrad	1	Computer Science	Yes
S14	F	Grad	2	Nursing	No
S15	F	Undergrad	2	Biology	Yes
S16	F	Undergrad	2	Agricultural Business Management	Yes
S17	M	Undergrad	3	Sociology	Yes
S18	F	Undergrad	1	Food Science & Life Science Communication	Yes
S19	F	Grad	1	Atmospheric and Oceanic Sciences	Yes
S20	M	Undergrad	4	Political Science	Yes
S21	M	Grad	1	Computer Science	Yes
S22	M	Undergrad	4	Mechanical Engineering	Yes
S23	F	Grad	1	Psychology	Yes
S24	F	Grad	1	Economics	No

Table 2: Instructors' Demographics in the formative study

ID	Department	Position	Teaching Experience	Class Size
I1	Public Affairs	Lecturer	1-2 years	30
I2	Business	Lecturer	6-9 years	100
I3	Chemistry	Lecturer	10+ years	200
I4	Mechanical Engineering	Assistant Professor	1-2 years	41
I5	Information Science	Teaching Faculty	3-5 years	15
I6	Biology	Teaching Faculty	10+ years	20
I7	Industrial Engineering	Assistant Professor	6-9 years	30
I8	Nuclear Engineering	Assistant Professor	6-9 years	30
I9	Biochemical Engineering	Research Analyst	6-9 years	108
I10	Computer Science	Teaching Faculty	10+ years	200
I11	English & Law	Associate Professor	10+ years	300
I12	Pharmacy	Teaching Faculty	6-9 years	50

fully understand something? Can you describe some specific examples?"), their confusion resolution strategies (*e.g.,* "*When your questions or difficulties don't get resolved immediately, how do you usually handle them?*"), the effectiveness and limitations of their current strategies (*e.g.,* "*How effective do you find this approach?*", "*Are there any drawbacks for you when using this method?*"), and their expectations for potential AI-powered learning support (*e.g.,* "*Imagine that there is a real-time answering tutor available during your class, providing instant answers or clarifications. How helpful do you think this tool would be? Why?*"). Instructors' questions focused on their challenges in gauging student understanding (*e.g.,* "*What is your experience with students asking questions during lectures?*"), current approach to addressing student questions (*e.g.,* "*What do you currently do to address student confusion?*", "*How well does this approach work?*"). After discussing these existing practices did we introduce the AI vignette and potential integration to understand instructors' expectations.

3.1.3 Data Analysis. To anonymize participants, we assigned IDs based on the order of study completion (*e.g.,* S1 refers to the first student participant, I1 refers to the first instructor participant). For qualitative data, we conducted a reflexive *Thematic Analysis (TA)* following Clarke and Braun [18] and McDonald et al. [47] to identify patterns and understand student and instructor perspectives. The first author and the third author, experienced in qualitative coding, transcribed and familiarized themselves with the data from audio recordings. Initial semantic codes were generated by the first author and organized into categories as the initial codebook. The team collaboratively coded data from the first participant using the initial codebook, discussed interpretations, and created a consensus-based codebook. Using the codebook, the remaining data were independently coded by the two authors, with peer reviews and iterative discussions to finalize codes. We constructed final themes from recurring, meaningful patterns in the data.

3.2 Findings

Through interviews with students and instructors, we identified five central themes reflecting the challenges and needs of students and instructors in lecture settings. These findings serve as foundations for deriving design implications, guiding the development of AI-powered systems that can more effectively mediate real-time learning support and instructional feedback.

3.2.1 Challenges in Following and Retaining Lecture Content. Many students described experiencing confusion during lectures for a variety of reasons. For many, the fast pace or density of the course material was difficult to follow (S6, S7, S18, S21, S22). For example, S6 stated, "*Sometimes professors rush through the content and I can't follow.*" Furthermore, several students noted that confusion often arose when instructors introduced complex concepts without providing sufficient elaboration or examples (S3, S5, S10). As S10 explained, "*Sometimes in organic chemistry he'll bring up a new concept that is pretty confusing because there're a lot of moving parts.*" Language and cultural barriers were particularly salient for international students (S2, S3, S8). For example, S2 stated, "*For the statistics class, I learned the concepts in Chinese before, but learning them in English is totally new.*" In addition, S8 added, "*When certain references are local, it is a bit challenging from time to time to grasp what this localized knowledge is assuming*". Besides in maintaining focus during long lectures, students also face challenges in recovering when they missed part of the lecture (S2, S11, S16–S22). Several participants noted that missing a key point often caused them to fall behind and struggle to catch up, leading to cascading confusion throughout the lecture. As S21 described, "*When there are times when I get behind in a lecture. Then I get stuck on it and then I fall behind in the rest of lecture.*"

Design Implication #1: These findings suggest the need for a system that can provide immediate clarification to help students recover from momentary lapses and support continuous participation during class.

3.2.2 Barriers to Asking Questions During Large Lectures. Many students expressed reluctance to ask questions during class, particularly in large lecture settings. Social anxiety (S17, S19, S23), fear of judgment (S9, S13, S16, S18, S21, S22), and concern about interrupting the lecture (S1, S3, S8, S17, S19, S20, S21, S23) were common deterrents. For instance, S16 expressed concerns "*to be that person to ask a stupid question,*" while S1 felt like "*it's rude to interrupt the whole class just for me.*" Several students felt isolated in their confusion and assumed others understood the material (S3, S10, S15). For example, S3 felt like "*the only one who has the problem here.*" Others referenced cultural or language barriers that made it harder to speak up (S2, S3, S8, S24): "*I'm worried about my oral English* (S24)." The issue was compounded in large classrooms, where the substantial number of the students discouraged active participation. As S13 expressed, "*it can be a little daunting to ask questions in class because there's so many other people in the room.*" Conversely, students reported greater comfort asking questions in smaller or discussion-based classes, or when they had built familiarity with the instructor (S4, S7, S8, S10, S16, S17, S20).

Design Implication #2: These results suggest that our system should be able to reduce the social costs of participation by enabling private, low-friction question submission.

3.2.3 Limitations in Student Strategies for Confusion Resolution. Students employed a variety of strategies to resolve confusion, including asking peers, searching online, reviewing lecture recordings, and attending office hours. AI tools like ChatGPT were also commonly used.

However, each of these methods had limitations. Peers were sometimes helpful but not always accurate: "*When I asked my friends, I feel less embarrassed. But it could almost be more confusing* (S16)." Online searches and videos were often time-consuming (S2, S9, S15, S16, S22): "*Searching online takes more time* (S2)," and "*YouTube videos are less effective*" sometimes if it is not the right video (S16). Students noted that ChatGPT was convenient but could produce vague or incorrect answers (S3, S14, S20, S21). For instance, S14 said that "*It tends to provide some general generic answers sometimes,*" while S3 mentioned that "*sometimes ChatGPT gives false ideas, so I double check with the professor.*"

Attending office hours was frequently cited as helpful but impractical due to scheduling conflicts (S4, S10, S11, S16, S20, S22), distance between home and school (S4, S12, S20, S22), or social anxiety (S17, S19, S23). For example, S22 shared that "*office hours have very set times...if I'm far from campus...I can't really make it.*"

Design Implication #3: These limitations underscore the need for an accessible and context-aware support system that minimizes effort and delay in resolving confusion during lectures.

3.2.4 Student Expectations for an AI-Powered Support Tool. Students articulated a set of desired features for a classroom AI assistant, including real-time transcription (S3, S17, S18, S22) and summarization of lecture content (S9, S13, S15, S17, S20, S23). S13 said that "*it would be nice if it could give me a summary of what the professor said for the past five or ten minutes* (S13)." They also wanted the ability to ask questions and receive real-time, concise, and contextually relevant answers (S7, S15, S18). For example, S18 said that it would be nice "*if it knows what's going on and I can ask it questions.*" Several students expressed interest in additional features such as generating practice questions (S10, S16, S17), summarizing key terms (S23), and explaining how different concepts connect (S21, S23).

In addition to describing desired functionalities, students also highlighted expectations for how the tool should fit into existing classroom practices. Many preferred a lightweight, low-friction interface that runs directly on devices they already use, such as laptops or tablets. They explained that such a tool could "*just split screen and have it in the corner, ready and available*" (S15), and that a website "*linked to Canvas*" aligns well with current workflows since "*most people already have their laptops out*" (S19). Several also favored using their own devices rather than carrying additional hardware, noting they "*didn't have to also have another device open*" (S18).

Design Implication #4: The tool should be seamlessly integrated into students' workflow, preferably accessible on devices they already use, such as laptops or phones, and should not require additional setup or effort during class.

3.2.5 Instructor Needs and Opportunities for AI Support. Instructors identified various needs and concerns regarding classroom dynamics and the potential integration of an AI assistant. First, instructors confirmed that students were often apprehensive about asking questions in large classes. Despite encouragement, "*students are very apprehensive of asking questions in large classes, and so it doesn't matter how often you try and invite answers or questions, no one asks* (I2)." I10 similarly observed, "*typically the students who are asking the questions are the most confident...the students who are the least confident often sit in the back and don't want to voice anything.*" Second, instructors found it difficult to accurately track student confusion during lectures, lacking real-time indicators of student understanding. As I11 explained, *It's very hard to tell how many students have already fully understood the concept...I have no way of measuring how and to what extent students are not understanding, except for exams.*" Similarly, I4 shared, "*I don't feel like I have indicators on whether the students actually know what I'm talking about.*" Some instructors had attempted to collect student questions during class using simple tools but yielded low participation (I2, I8). For example, I2 "*tried Google Sheets...but students don't seem to be interested in filling in questions during lecture.*"

Therefore, instructors expressed strong interest in having a tool that not only encourages students to ask questions, but also provides a safe, anonymous channel for doing so. A key reason behind students' reluctance was their concern about privacy and fear of judgment from peers. Several instructors emphasized that anonymity would be critical to making students feel comfortable asking questions (I2, I5, I10). As I2 explained, "*I would like everything to be anonymized in some way, shape or form so that a student can ask questions and not have to worry about other students knowing who is asking the question.*"

Beyond facilitating anonymous question submission, instructors also hoped that an AI assistant could help respond to student questions in real time. However, they raised concerns regarding the accuracy and appropriateness of AI-generated answers (I4, I6–I8, I12). Instructors emphasized the importance of ensuring the consistence of responses from AI with the course content and their own teaching style (I1, I10, I12). As I12 highlighted, "*I would want to have a chance to look at the type of content that is being sent to students.*" Consequently, instructors wanted the ability to review and monitor AI responses to maintain quality and consistency.

In addition, instructors want to know what questions students were asking during class. They wanted access to both raw questions (I7, I8) and aggregated questions grouped by topics (I2, I3, I5–I8, I10, I12). For example, I7 mentioned "*based on my current experience, almost all questions are important questions.*" Other instructors emphasized the importance of grouping similar questions by topics or learning objectives to highlight common areas of confusion. For instance, I2 noted "*I'd like some sort of grouping and ranking of problems so I know which topic was generating the most questions.*" To utilize the questions in real-time, instructors favored simple, aggregated displays like top-question lists. As explained by I5, "*being able to have an aggregated summary list of the top five questions, but then also seeing the individual questions because it may be complex or may be a little nuanced.*" They stressed that interfaces must be quick to check and minimally disruptive (I2, I4, I6, I8–I11).

Finally, instructors saw great potential for such a system that can help them reflect on and improve their teaching. Reviewing patterns in student confusion could help adjust explanations and lecture pacing. As I6 stated, "*If I've got the most common types of questions for a particular lecture, I can use that information when I plan that lecture for the next time.*"

Design Implication #5: These insights suggest that an effective AI-powered classroom assistant should not only support real-time clarification for students, but also provide actionable feedback to instructors to enhance teaching effectiveness.

4 AskNow

Based on the formative study findings, we designed and developed AskNow, an interactive system powered by a large language model (LLM) to support real-time question answering in large-scale classrooms. AskNow consisted of a student and an instructor interfaces. We show the architecture of AskNow in Figure 2.

We carefully structured AskNow's functionality based on the explicit challenges and requirements identified from our formative study. In particular, each core feature directly addresses the five Design Implications (DIs #1–#5) we derived in Section 3. Students reported significant difficulties in following fast-paced lectures, language and cultural barriers causing comprehension issues, and hesitancy in public questioning due to social anxiety. Conversely, instructors expressed challenges in gauging real-time student comprehension, and wanted effective tools to detect and address areas of widespread confusion promptly. In response, AskNow incorporates three core phases to directly address these identified needs: (1) real-time lecture transcription, (2) context-aware question answering, and (3) question aggregation.

4.1 Real-time Lecture Transcription

On the instructor interface, AskNow provides real-time, automatic speech-to-text transcription of lecture audio. This real-time transcription functionality not only provides instructors with a comprehensive textual record of their lecture but also creates an essential context data for the language model, serving subsequent functionalities such as context-aware question answering. This ensures that AskNow-generated responses are precise, contextually relevant, and directly linked to the ongoing lecture content, which directly addresses Design Implication #1 by offering students immediate clarifications to help them catch up on missed content and supports Design Implication #3 through continuous real-time availability of contextual transcript information.

4.2 Context-aware Question Answering

On the student interface, AskNow allows students to privately submit questions and obtain real-time, contextually accurate responses,

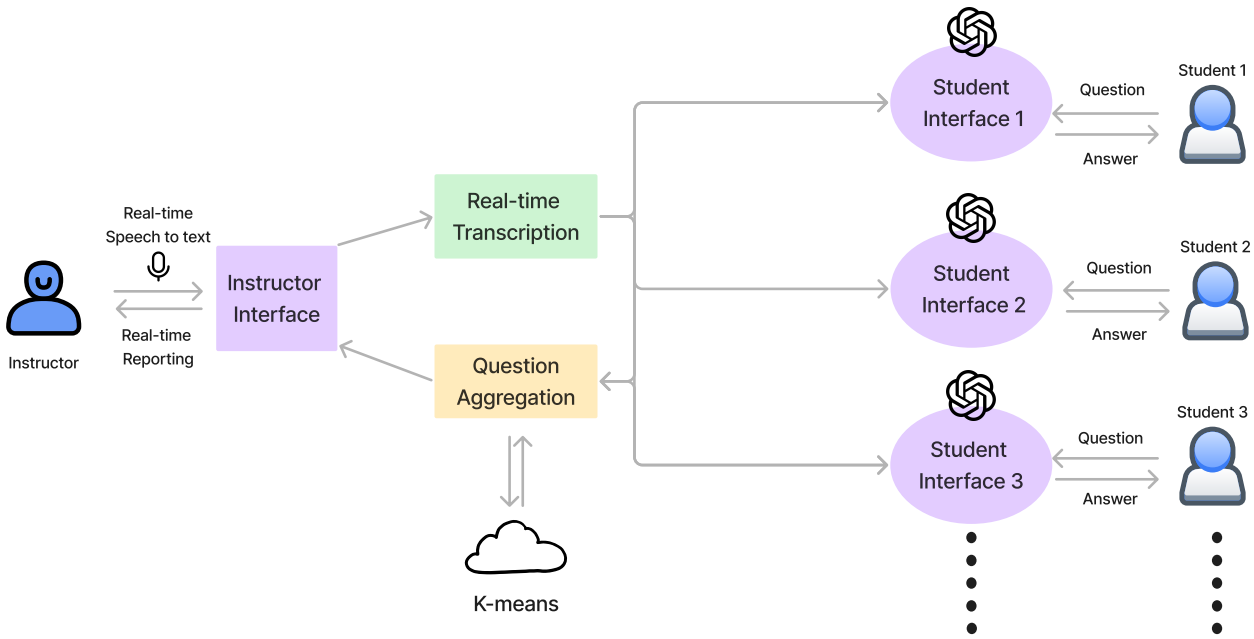


Figure 2: AskNow system architecture and end-to-end data flow. Student interfaces send questions to a cloud backend, which retrieves the ongoing lecture transcript (streamed from the Instructor Interface via real-time speech-to-text) and lecture metadata to construct a context-aware prompt for the LLM; the generated answer is then returned to the originating student. In parallel, all question events are logged, embedded, and clustered into topic groups that are displayed on the Instructor Interface to surface common areas of confusion in real time.

addressing the barriers identified in our formative study, such as social anxiety, fear of interrupting lectures, and concerns about judgment (Section 3.2.2). Aligned with Design Implication #2, students can ask questions anonymously, reducing social costs and hesitation that often hinder large-class participation. Furthermore, following Design Implication #3 and #4, the student interface is designed in a GPT-like chat format to minimize the learning curve; users can seamlessly interact with the system in a familiar, conversational style without additional setup or instruction, ensuring minimal disruption to students' existing workflow and allowing them to use familiar devices (laptops, phones) in class. Specifically, when a student submits a question, the system retrieves the current lecture transcript to ensure LLM responds based on the current classroom context. The whole transcript is divided into 30-second segments and added corresponding timestamps to enable context referencing on temporal information (e.g., "What did the instructor say ten minutes ago?"). See Figure 3 for an example. This segmentation is used only for storage and indexing, which does not limit the amount of context provided to the LLM. This design directly addresses Design Implication #1, which highlights the need for immediate clarification and recovery from missed or confusing lecture segments.

To further enhance the precision and relevance of the generated responses, instructors provide lecture-specific metadata—including lecture title, subject, and instructor name—upon initiating the lecture session. This metadata is incorporated directly into the prompt sent to the OpenAI GPT-4o model, aiding it in accurately interpreting lecture contexts and generating responses that align closely with course content and instructional style.

Moreover, the prompt engineering strategy includes carefully crafted instructions to ensure appropriateness, safety, and consistency in the generated responses. For example, the prompt explicitly instructs the model to always reference the lecture transcript first, maintain a neutral and informative tone, and adhere strictly to academic and technical accuracy. Additional safety guidelines prevent misuse or unintended behavior, instructing the model to decline engaging with inappropriate requests and ensuring compliance with content policies. Please refer to Appendix Section A for the full system prompt.

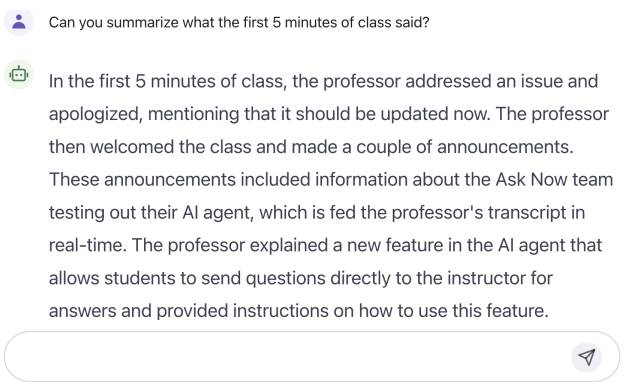


Figure 3: An example of AskNow responding to a time-relevant question through the student interface, where the student asked the LLM to summarize the first five minutes of the lecture.

Finally, in selecting the LLM to power this context-aware answering module, we conducted careful comparisons among various advanced models, including Anthropic's Claude 3.5 Haiku, GPT-4o-mini, and GPT-4o. We ultimately selected GPT-4o based on its superior balance of response accuracy and cost efficiency, which are particularly critical for scalability and effectiveness in large-scale classroom environments.

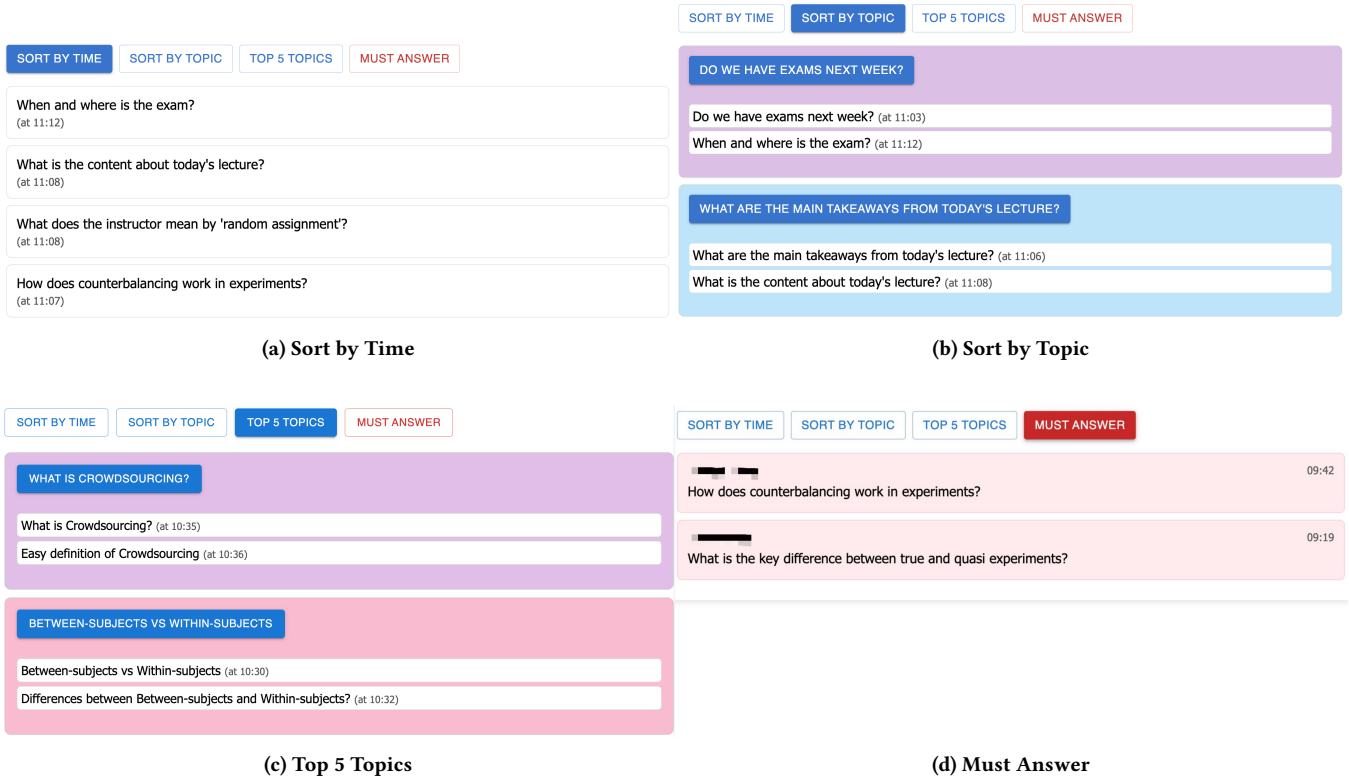
4.3 Question aggregation

We used multiple strategies, including embedding-based clustering, to aggregate student questions and provide instructors with a concise yet comprehensive view of common misunderstandings. Once students submit their questions through the AskNow interface, these are automatically processed and displayed on the instructor-facing web dashboard. Aligned with Design Implication #5, this approach offers real-time, actionable insights into student confusion. Drawing on our formative study (Section 3.2.5), we implemented four distinct methods for organizing and presenting questions, and the UI design is presented in Figure 4:

1. Sort by Time. All submitted questions are displayed in reverse chronological order, prioritizing recent student queries at the top of the instructor interface. This approach helps instructors quickly identify emerging points of confusion in real-time during the lecture.

2. Sort by Topic. To facilitate concise and efficient comprehension of complex growing student inquiries, we embedded each question into a semantic vector space. Each question is preprocessed into a high-dimensional embedding vector that captures its semantic content. Leveraging these embeddings, we performed clustering by calculating pairwise Euclidean distances between vectors and recursively applying a threshold-based, K-means algorithm that does not require a fixed K. When a new question arrives, it is assigned to the existing cluster whose centroid is within a predefined distance threshold, or a new cluster is created if no such cluster exists. This predefined distance threshold, selected through an iterative testing to balance cluster granularity and stability, ensures that questions are clustered based on actual semantic similarity rather than arbitrary or predefined group sizes, thus maintaining consistent clustering accuracy regardless of class size variations. For each cluster, we use the question whose embedding lies closest to the cluster centroid as the representative "topic question," and display this question as the title of the block in the instructor interface. After clustering, the resulting topic-based question groups are sorted by the most recent timestamp, allowing instructors to easily identify and respond to current, collective student misunderstandings.

3. Top 5 Topics. Due to instructors' expressed need to swiftly identify and address the most critical and common questions during a lecture, we provided a streamlined "Top 5 topics" view showcasing only the top five most frequently occurring question clusters. To maximize relevance and minimize noise, we applied heuristic filters excluding overly general or non-productive queries, such as requests to "catch me up," greetings ("Hello AskNow"), overly broad summaries of lecture content, or extremely short questions.



(a) Sort by Time

(b) Sort by Topic

(c) Top 5 Topics

(d) Must Answer

Figure 4: Instructor interface with four modes of question aggregation: (a) Displays recent questions at the top; (b) Clusters semantically similar questions; (c) Highlights the most frequent clusters; (d) Shows questions flagged by students for instructor clarification. Screenshots are taken from the actual deployment of the system. In (d), student names are blurred to preserve anonymity.

This targeted approach helps instructors efficiently prioritize areas needing immediate attention without overwhelming them with redundant or low-value queries.

4. Must Answer. Recognizing the potential limitations of AI-generated responses—particularly as noted by students who sometimes require direct instructor clarification—we introduced a special interaction mode. After receiving AI-generated responses, students may use a dedicated "Send to Instructor for Answer" button if they feel their question remains unresolved or unsatisfactorily answered. Such questions, analogous to traditional hand-raising, are separately collected under the "Must Answer" category, prominently displayed to ensure instructors promptly address these pressing concerns.

These four modes of aggregation collectively provide instructors with actionable, real-time insights into students' shared points of confusion. This design not only addresses the need for scalable monitoring but also ensures that critical questions requiring direct instructor input remain visible and prioritized.

4.4 System Implementation and Technical Details

AskNow is designed as an interactive system with an architecture tailored to the dynamic, high-concurrency, and low-latency requirements of real-time classroom use. In this section, we report the main technical challenges and the corresponding implementation strategies.

4.4.1 Dynamic, Low-Latency Context Management. To precisely understand and utilize a constantly changing lecture context in real time, we developed a decoupled full-stack architecture. The student-facing interface is implemented in Next.js (TypeScript) and deployed on Vercel for elastic, Internet-facing scale, while the instructor interface is hosted privately on cloud servers for secure, real-time monitoring. A Node.js + Express backend on AWS EC2 [1] provides a reliable infrastructure for concurrent requests, with Socket.IO handling real-time bidirectional communication.

We implemented efficient context retrieval through differential updates strategy. Returned transcripts are incrementally materialized on the server as structured, segmented JSON with stable segment indices and timestamps. When answering, it fetches only

[1] https://aws.amazon.com/ec2/

the newest transcript segments necessary for context, thereby reducing token costs, avoiding redundant context, and keeping end-to-end latency low. Explicit segmentation also supports traceability of responses against specific transcript spans.

In addition, we integrate AssemblyAI's real-time Speech-to-Text API [2] to support uninterrupted streaming transcription. Its ability to handle indefinite-duration sessions prevents context loss or latency that would otherwise occur from session renewals during long lectures. To balance semantic completeness with responsiveness, lecture transcripts are segmented into 30-second windows, a trade-off identified through iterative testing. Shorter windows risk capturing incomplete semantic units, while longer ones dilute the context for time-sensitive questions and introduce unacceptable end-to-end latency. These windows serve as the basic units for incremental transcript storage and timestamping, whereas the context retrieved for question answering may span all accumulated windows since the beginning of the lecture.

4.4.2 High-Concurrency, Real-time Semantic Aggregation. To convert the high volume of incoming student questions into an interpretable dashboard for instructors, we implemented a multi-stage aggregation pipeline. A threshold-based, recursive K-means algorithm groups semantically similar questions without forcing them into arbitrary cluster sizes. This approach maintains high semantic coherence even under rapid increases in question volume and ensures clusters remain stable, giving instructors a consistent real-time view. The distance threshold can also be configured to adjust the granularity of generated topics.

Our technology choices were performance-driven. We deliberately avoided computationally intensive alternatives, such as running local embedding models or employing LLMs for dynamic summarization or classification, as these would introduce unacceptable latency, disrupt transcript storage and retrieval, and consume unsustainable levels of API tokens in large classrooms. Instead, the aggregation path is engineered to remain lightweight and non-blocking. We selected OpenAI's `text-embedding-3-small` [3] API, which offers a favorable balance of embedding quality, speed, and cost efficiency. This design enables accurate, contextually relevant aggregation while maintaining low latency and system responsiveness when many students interact simultaneously, ensuring that instructors receive timely and reliable insights into student needs. Additional implementation details—including the full system prompt, transcript segmentation configuration, example transcript records, question logging schema, and pinned model and API versions—are provided in Appendices A–E to support reproducibility of our implementation.

5 Evaluation

To evaluate the effectiveness and usability of AskNow in real classroom settings, we conducted a series of studies to answer three research questions:

- (1) How effectively does AskNow address student questions?
- (2) How do students use and perceive AskNow?
- (3) How do instructors use and perceive AskNow?

[2] https://www.assemblyai.com/products/streaming-speech-to-text
[3] https://platform.openai.com/docs/models/text-embedding-3-small

5.1 Participants

We recruited participants from three university-level courses offered by the Department of Computer Sciences at the University of Wisconsin–Madison: an introductory undergraduate course in web development, an intermediate undergraduate course in building user interfaces, and an advanced graduate-level course in human–computer interaction.

In total, 88 students were enrolled in the introductory course, 129 in the intermediate course, and 69 in the advanced course. Among them, 51, 73, and 48 students, respectively, signed up for the study. Students who completed all study components—including the pre-survey, lecture attendance, system use, and post-survey—numbered 32, 48, and 37, respectively, across the three courses. Table 3 summarizes the three courses and the participation of students.

Table 3: Summary of student participation across courses.

Course ID	Course Level	Class Size	Signed Up	Completed Study
C1	Introductory	88	51	32
C2	Intermediate	129	73	48
C3	Advanced	69	48	37

Additionally, three course instructors participated in the study by evaluating student questions and AskNow's responses, and by participating in post-study interviews.

5.2 Study Sessions and Procedure

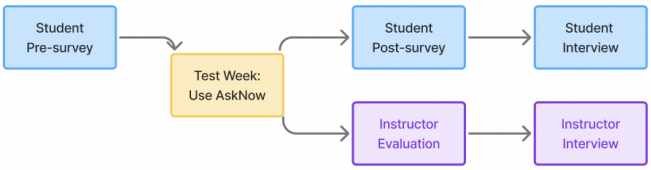


Figure 5: Study Procedure

We obtained IRB approval for the study and coordinated with the course instructors for participant recruitment. Two weeks before the deployment, we invited students to participate through oral announcements during lectures and notifications posted on each course's Canvas page. Participation was entirely voluntary. Figure 5 demonstrates the study sessions and procedure.

Students who agreed to participate were required to complete three steps:

(1) Pre-survey: A 5-minute online survey that collected demographic information and assessed students' perceived course understanding, frequency of confusion, and the current efficiency of resolving confusion during lectures.
(2) System use: Attending the two in-person lectures during the test week, during which students were provided access to AskNow and could use it as needed to ask questions and receive real-time answers.

(3) Post-survey: A 10-minute online survey administered after the test week, measuring changes in perceived course understanding, information retention, frequency of confusion, and efficiency in resolving confusion after using AskNow. Participants also completed a USE (Usefulness, Satisfaction, and Ease of Use) questionnaire [45].

Students who completed all three steps were awarded 1 extra credit point toward their final course grade. At the end of the post-survey, participants were asked whether they were willing to participate in a follow-up interview. From the pool of volunteers, we randomly selected eight students from each course (24 students in total) to participate in a 30-minute semi-structured interview. Students who completed the interview received a $10 compensation.

In addition to student feedback, we collected system interaction logs, comprising the questions students asked and the corresponding answers generated by AskNow. Course instructors later evaluated a sample of these question–answer pairs and then participated in a 30-minute semi-structured interview to reflect on their use of AskNow in the classroom and on the quality of its responses (see Section 5.3.1 for details of the expert evaluation procedure). Each instructor was compensated $50 for their time.

5.3 Data Analysis

To comprehensively assess the effectiveness and usability of AskNow, we conducted a multi-method analysis that combined expert validation of system responses, quantitative comparisons of pre- and post-survey results, and qualitative analysis of student and instructor interviews. Below, we detail the methods used for each analysis component.

5.3.1 AskNow Response Validation. To evaluate the accuracy and perceived quality of AskNow's answers, we performed expert validation using instructors' ratings. For each course, we randomly sampled 100 question-answer pairs generated during the study sessions. Instructors were asked to independently assess each pair along two dimensions: `perceived correctness` and `perceived satisfaction`, both rated on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Because instructors evaluated 100 question–answer pairs per course, we adopted a shorter 5-point scale to reduce their rating burden while still capturing meaningful variation in their judgments.

We calculated descriptive statistics for both dimensions across each course. These ratings provided an expert-grounded evaluation of the reliability and relevance of AskNow's responses within the context of real classroom use.

5.3.2 Quantitative Analysis. Quantitative analyses focused on comparing students' self-reported learning experiences before and after using AskNow.

We administered matched pre- and post-surveys to assess four perceived dimensions: (1) overall understanding of course material, (2) information retention from lectures, (3) frequency of confusion during lectures, and (4) time required to resolve confusion when confused. Responses were collected using 7-point Likert scales (1 = strongly disagree, 7 = strongly agree). To maintain consistency across student-facing measures, and in line with the standard USE questionnaire [45], all student survey items in our study–including both the pre/post learning items and the USE scales–used 7-point response options.

For each dimension and within each course, we conducted paired-sample t-tests comparing pre- and post-survey scores. Because we ran four related tests per course, we treated them as a family of comparisons and applied Bonferroni correction to control the family-wise Type I error rate [71], adjusting the per-test significance threshold to $\alpha_{\text{adj}} = .0125$. Following current reporting recommendations [72], we report both unadjusted and Bonferroni-corrected p-values, along with standardized effect sizes (Cohen's d for repeated measures) and 95% confidence intervals for the mean pre–post differences (Table 4). For interpretation, we use Cohen's conventional benchmarks for d (0.2, 0.5, and 0.8 indicating small, medium, and large effects, respectively) [19].

Additionally, we analyzed responses to the USE (Usefulness, Satisfaction, and Ease of Use) questionnaire administered in the post-survey. We computed mean ratings for each of the four subscales—*Usefulness*, *Ease of Use*, *Ease of Learning*, and *Satisfaction*—to assess students' perceptions of AskNow's usability.

5.3.3 Qualitative Analysis. We conducted thematic analysis on semi-structured interview data collected from 24 students and 3 instructors.

All interviews were audio-recorded, transcribed verbatim, and analyzed using an inductive thematic coding approach. Two researchers independently reviewed the transcripts, generated initial codes, and iteratively refined emerging themes through discussion and consensus-building. We used the same analysis methods as we did in 3.1.3. In this phase, we similarly assigned IDs based on the order of study completion (*e.g.,* P1 refers to the first student participant, In1 refers to the first instructor participant).

The qualitative findings complement our quantitative analyses by offering nuanced insights into users' experiences and perceptions, highlighting the strengths of AskNow as well as areas for further system refinement.

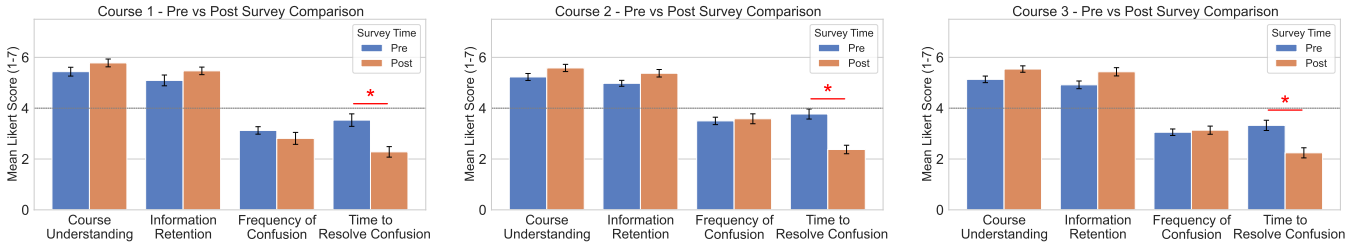
6 Results

We present the results of our evaluation organized around the three research questions. We first report on the effectiveness of AskNow in supporting student learning, followed by an analysis of usage patterns, and finally, a thematic analysis of students' and instructors' perceptions.

6.1 Effectiveness of AskNow

We assessed the effectiveness of AskNow through three perspectives: students' perceived learning experience before and after using the system, instructors' evaluations of the system-generated responses, and students' perceived usability based on the USE questionnaire [45].

6.1.1 Pre- and Post-Survey Results. Among the four perceived learning dimensions, students' ratings of the time required to resolve confusion during lectures showed the clearest and most robust pre–post changes. Across all three courses, mean scores on this item decreased by roughly 1.1–1.4 scale points on the 7-point Likert scale (lower scores indicate shorter perceived time), and these differences remained statistically significant after Bonferroni correction



(a) C1 (Introductory) **(b) C2 (Intermediate)** **(c) C3 (Advanced)**

Figure 6: Pre- and post-survey comparisons of students' perceived learning experience across three courses: (a) Course 1 (Introductory), (b) Course 2 (Intermediate), and (c) Course 3 (Advanced). The y-axis represents mean scores on a 7-point Likert scale; error bars represent ±1 standard error of the mean (SEM). Asterisks indicate pre–post differences that remain statistically significant after Bonferroni correction for the four tests per course.

for the four tests per course (all $p_{\text{adj}} < .0125$; Table 4). The corresponding effect sizes were in the medium range ($|d| = 0.62–0.77$), and the 95% confidence intervals for the mean differences did not include zero. Taken together, these results indicate that students perceived AskNow as meaningfully accelerating how quickly they could resolve confusion in class.

For overall understanding of course material and information retention from lectures, we observed small-to-medium positive changes, particularly in the intermediate and advanced courses. For example, in Course 3, students' perceived understanding and retention increased by approximately 0.4–0.5 points on the 7-point scale (see Table 4). Several of these contrasts reached conventional significance before correction (e.g., $0.02 \leq p_{\text{raw}} \leq 0.08$), but none remained significant under the Bonferroni-adjusted threshold (all $p_{\text{adj}} \geq .09$). We therefore interpret these improvements as promising trends rather than confirmatory evidence of changes in these broader aspects of perceived learning.

By contrast, we did not observe systematic pre–post differences in students' self-reported frequency of confusion during lectures (all $|d| \leq 0.22$, all $p_{\text{adj}} \geq .89$).

Detailed descriptive statistics, confidence intervals, and effect sizes for each item and course are reported in Table 4 and visualized in Figure 6, where asterisks indicate only those pre–post differences that remain significant after Bonferroni correction.

In addition to these changes in students' perceived learning experience, we evaluated the quality and usability of AskNow from both instructor and student perspectives.

6.1.2 Instructor Evaluation of Response Quality. To assess the perceived quality of AskNow's responses, course instructors were invited to rate randomly sampled question-answer pairs on two dimensions: *perceived correctness* and *satisfaction* (both on a 5-point Likert scale). Across all three courses, instructors consistently rated the responses highly, indicating that AskNow provided accurate and satisfactory answers overall.

Specifically, the perceived correctness scores, measured on a 5-point Likert scale, were 4.88 (Course 1), 4.79 (Course 2), and 4.53 (Course 3). Satisfaction ratings were similarly high, with scores of 4.75 (Course 1), 4.51 (Course 2), and 4.45 (Course 3). These results suggest that instructors found AskNow to be a reliable support tool for addressing student questions during lectures. A summary of instructors' ratings across the three courses is illustrated in Figure 7.

6.1.3 Usability Evaluation (USE Questionnaire). We further administered the standard USE (Usefulness, Satisfaction, and Ease of Use) questionnaire to students who used AskNow during the study. Students' responses were highly positive across all dimensions.

Across the three courses, average ratings for *Usefulness*, *Ease of Use*, *Ease of Learning*, and *Satisfaction* were all above 5 out of 7 (Table 5). Notably, *Ease of Learning* received exceptionally high ratings, exceeding 6 in all three courses, indicating that students found the system intuitive and easy to adopt with minimal effort.

These results demonstrate that AskNow was not only effective in supporting students' learning during lectures but was also well-received in terms of usability and user satisfaction.

6.2 Usage Patterns

Across the three participating courses, students generated a total of 302 (Course 1), 438 (Course 2), and 321 (Course 3) questions through the AskNow system. We first filtered out irrelevant questions—such as greetings (*e.g.*, "*Hello,*" "*How are you?*") and inquiries unrelated to

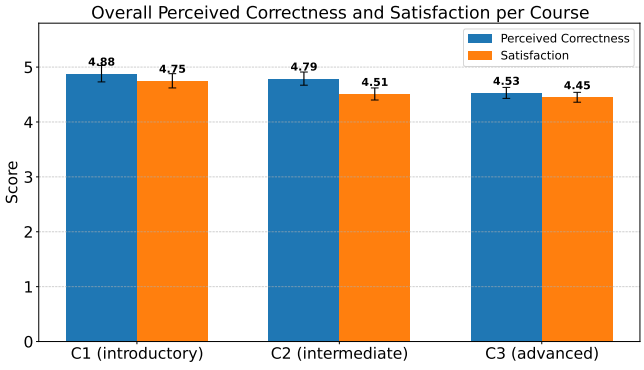


Figure 7: Instructor-rated correctness and satisfaction across three courses. Error bars represent ±1 standard error of the mean (SEM). Instructors in all three courses gave high ratings on the perceived correctness and satisfaction of AskNow.

Table 4: Pre–post descriptive statistics, effect sizes, and p-values for perceived learning measures by course. Means and standard deviations are based on paired responses. Positive ΔM values indicate higher post-survey ratings; negative values on "Time to resolve confusion" indicate shorter perceived time.

Course	Dimension	n	Pre M (SD)	Post M (SD)	ΔM [95% CI]	d	p_{raw}	p_{adj}
Course 1	Course understanding	32	5.44 (0.98)	5.78 (0.87)	+0.34 [-0.04, 0.73]	0.32	.078	.311
Course 1	Information retention	32	5.09 (1.20)	5.47 (0.84)	+0.38 [-0.12, 0.87]	0.27	.136	.543
Course 1	Frequency of confusion	32	3.12 (0.83)	2.81 (1.33)	-0.31 [-0.83, 0.20]	-0.22	.224	.895
Course 1	Time to resolve confusion	32	3.53 (1.39)	2.28 (1.17)	-1.25 [-1.97, -0.53]	-0.62	.001	.005
Course 2	Course understanding	48	5.23 (0.95)	5.58 (0.99)	+0.35 [-0.03, 0.74]	0.27	.071	.284
Course 2	Information retention	48	4.98 (0.81)	5.38 (1.02)	+0.40 [0.02, 0.77]	0.30	.040	.161
Course 2	Frequency of confusion	48	3.50 (1.01)	3.58 (1.35)	+0.08 [-0.40, 0.57]	0.05	.730	1.000
Course 2	Time to resolve confusion	48	3.77 (1.37)	2.38 (1.16)	-1.40 [-1.92, -0.87]	-0.77	$< .001$	$< .001$
Course 3	Course understanding	37	5.14 (0.79)	5.54 (0.77)	+0.41 [0.06, 0.75]	0.39	.023	.093
Course 3	Information retention	37	4.92 (0.92)	5.43 (0.99)	+0.51 [0.06, 0.96]	0.38	.026	.104
Course 3	Frequency of confusion	37	3.05 (0.78)	3.14 (0.98)	+0.08 [-0.33, 0.49]	0.07	.692	1.000
Course 3	Time to resolve confusion	37	3.32 (1.23)	2.24 (1.21)	-1.08 [-1.63, -0.53]	-0.65	$< .001$	.001

the course material (*e.g.*, "*How's the weather today?*"). After filtering, 255 questions in Course 1, 417 in Course 2, and 204 in Course 3 remained for analysis.

The first and third authors independently coded all retained questions into five categories based on their primary intent:

- **Catch me up**: Questions aimed at real-time support to catch up with the ongoing lecture content. Examples include "*I just arrived to class. Can you quickly catch me up on what I've missed?*", "*Catch me up! What did the instructor just explain?*", and "*What was the student's question and instructor's answer?*".
- **Definition or Information**: Inquiries seeking clarification or further information about lecture-related concepts. Representative questions include "*Can you explain falsifiability?*", "*How does an independent variable affect the dependent variable during hypothesis formulation?*", and "*Does a quasi-experiment always include both pre- and post-tests?*".
- **Summarization**: Requests for summarizing the content covered in the lecture. Typical examples are "*Help me summarize the main topics covered today.*", "*Summarize the key points the instructor talked about.*", and "*What were the main ideas discussed so far in class?*".
- **External Knowledge**: Questions that required information beyond the immediate lecture context. Examples include "*Did the professor make any mistake?*", "*Where does crowdsourcing come from?*", and "*Are there examples of crowdsourcing applied in structural engineering?*".
- **Other Types**: Questions that did not fit into the above categories, such as inquiries about course logistics, technical troubleshooting, or tool usage. For instance, students asked "*Do we have a discussion post due tomorrow?*", "*How do I install Node.js and npm?*", "*Can you translate what the professor is saying in Hindi?*", and "*Why would I be getting an infinite fetch loop?*".

Table 5: USE questionnaire mean scores (1–7) by course and subscale.

Course	Usefulness	Ease of Use	Ease of Learning	Satisfaction
Course 1	5.57	5.80	6.07	5.70
Course 2	5.42	5.85	6.18	5.40
Course 3	5.17	5.61	6.09	5.10

The distribution of question types across the three courses is shown in Figure 8. Across all three courses, the majority of questions fell into the **Catch me up** and **Definition or Information** categories, collectively accounting for more than half of all questions in each course. This finding indicates that students predominantly used AskNow to seek clarification on lecture content and to reinforce their understanding during class.

Moreover, based on the total number of questions and the number of participants, we estimate that each student asked an average of 4 to 7 questions during the study sessions. This suggests that students were actively engaging with the system throughout the lectures.

Interestingly, in Course 2, a course with a strong focus on practical skills development, there was a noticeably higher proportion of **Other Type** questions, particularly those related to technical support and tool usage (*e.g.*, npm installation, Git commands). This observation highlights AskNow's potential to assist students not only in content comprehension but also in navigating technical challenges in hands-on learning environments.

6.3 Student Perception

Beyond the analysis of logged questions, we conducted thematic analysis on post-study interviews with 24 students to further understand their experiences and use of AskNow. This qualitative analysis complements the quantitative findings by offering deeper insights into students' usage behaviors, motivations, perceptions of the system, and suggestions for improvement.

6.3.1 Usage Contexts. Students reported using AskNow primarily to catch up on missed content during lectures, particularly when they were late, distracted, or engaged in parallel activities. For instance,

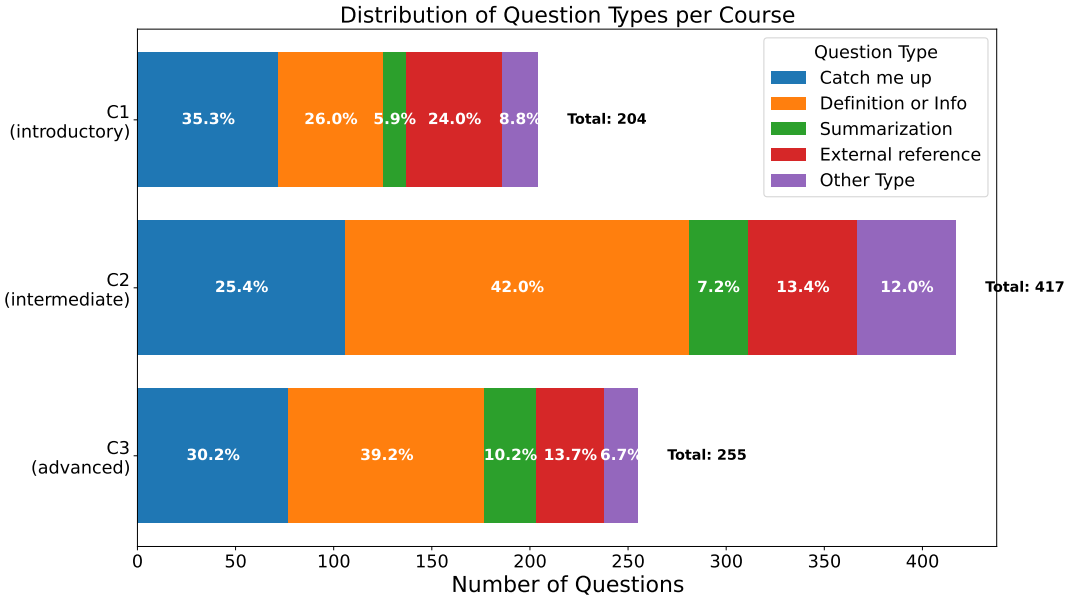


Figure 8: Distribution of question types across different courses

P6 noted, *"Maybe if I miss something, I'll just ask it to tell me what the professor has said"*, while P7 described, *"Usually I get late to two-hour courses... I would just use it to catch up with the lecture"*. Similarly, P9 shared, *"If I stopped paying attention for a period of time, I would ask what the professor had just said"*.

In addition to catching up, students also frequently used AskNow to request explanations for complex concepts encountered during lectures. P13 recalled, *"When the teacher introduced new concepts like useEffect, I used AskNow to help me understand it"*, and P21 explained, *"When I lose focus and don't understand a concept or keyword, I just ask for further explanation"*.

Other less common but noteworthy usage patterns included generating study guides after class (P1) and creating exercises for additional practice (P3). These findings suggest that students leveraged AskNow both for real-time clarification and for post-lecture reinforcement of learning.

6.3.2 Motivations for Using AskNow. Several motivations drove students to prefer AskNow over traditional in-class question-asking. A prominent motivation was the desire to avoid disrupting the lecture flow. P15 remarked, *"I don't want to interrupt the professor because it is not good and it makes the course not fluent"*, while P21 stated, *"Asking AskNow is great because you don't interrupt the lecture"*.

Students also emphasized the reduction of social pressure and anxiety when using AskNow. P13 shared, *"I'm really nervous about asking questions during the lecture, especially as an international student"*, and P17 noted, *"Sometimes we feel a bit shy to ask in front of the entire class, but AskNow allowed us to ask anonymously"*.

Efficiency was another frequently cited motivation. Students appreciated the immediacy and convenience of receiving answers without waiting. P6 explained, *"It gives me a very fast answer. I don't need to browse Google"*, and P21 highlighted, *"You don't need to interrupt, you don't need to raise your hand; no barrier"*.

These motivations collectively illustrate how AskNow addressed key barriers to help-seeking behaviors in large lecture environments, making real-time clarification more accessible and less intimidating.

6.3.3 Perceptions of AskNow's Responses. Overall, students perceived AskNow's responses as accurate, clear, and supportive of their learning. Multiple participants expressed trust in the correctness of the system's outputs. P1 remarked, *"None of the answers were wrong that I got, so I'm assuming that it was right"*. Others highlighted the readability and structure of the responses, with P7 stating, *"The chatbot gave me answers in bullet points. It was very straightforward to read"*.

Students who were non-native English speakers particularly valued the system's ability to provide explanations in their preferred language. P8 commented, *"Because my native language is not English, I can chat in my language and better understand what the teacher said"*, and P14 added, *"AskNow can help me understand phrases or sentences that I might have missed the first time"*.

These perceptions suggest that AskNow successfully delivered timely and comprehensible responses, enhancing students' understanding regardless of their language background.

6.3.4 Suggestions for Future Improvement. While students were generally satisfied with AskNow, they also identified several areas for improvement.

One frequently suggested enhancement was integrating additional course materials, such as lecture slides or notes, to provide more grounded and context-specific answers. As P1 suggested, *"Maybe allow AskNow to access professor slides or notes"*, and P8

proposed, *"Maybe it can keep chasing which slide the teacher is showing"*.

Students also expressed a desire for customizable response lengths to suit different needs. P17 proposed, *"Maybe have an option for a brief answer or a detailed one"*, while P21 noted, *"Some users want longer answers, but for non-native speakers, shorter responses might be better"*.

Additional suggestions included supporting uploads of images and PDFs (P4, P17), introducing shortcut buttons for common queries (P6), enabling after-class access for review purposes (P10), and providing options for user-controlled anonymity when submitting questions (P4).

These suggestions highlight students' aspirations for more flexible, personalized, and context-aware real-time learning support through systems like AskNow.

6.4 Instructor Perception

Thoughts about the AskNow Instructor-side Features. Instructors emphasized the importance of avoiding recording sensitive information about students to protect student privacy. All instructors appreciated the *Pause* feature, recognizing its importance during in-class activities or personal student interactions to avoid recording irrelevant or sensitive conversations. In1 highlighted that pausing could help maintain student privacy, aligning with FERPA regulations. In2 mentioned using it naturally during breaks and activities in class to avoid feeding unrelated content into AskNow.

Regarding the four ways to display student questions, instructors preferred using the interface primarily during breaks or after lectures rather than in real time. In1 and In2 favored post-lecture review to reflect on student confusions. In2 and In3 emphasized that categorizing questions by topics helped them quickly grasp areas of confusion, although In3 noted that many displayed questions asked for summaries, which were less useful to the instructor during the lecture.

Quality and Limitations of Responses. Overall, instructors found AskNow's responses to be accurate and helpful. In1 expressed high satisfaction, stating that only a small percentage of the responses had minor problems, mainly due to outdated knowledge. For example, AskNow suggested an outdated way to create a React project, which might mislead student. Similarly, In2 found responses to be generally well-aligned with the course content but observed that the system sometimes generated irrelevant code snippets when explaining high-level design concepts, which could potentially confuse students. In3 acknowledged the system's strength in summarization and explanation tasks but pointed out several limitations, including transcription errors affecting answer quality, inconsistencies in summarizations, and occasional hallucinations, particularly when the system attempted to reference visual materials it did not have access to.

Perceived Benefits. The instructors identified several perceived benefits of AskNow for students. They noted that the system provided real-time clarification, allowing students who arrived late or missed parts of the lecture to catch up effectively.

In2 and In3 emphasized that AskNow supported shy students who might otherwise hesitate to ask questions in a large lecture setting.

Furthermore, In1 noted that AI-assisted responses helped students identify small mistakes in in-class exercises, providing immediate feedback that enhanced their learning experience.

For instructors, AskNow offered opportunities for post-lecture reflection. In2 highlighted that the aggregated student questions could inform improvements in future course design by identifying areas of recurring confusion.

Concerns. Despite recognizing its benefits, instructors raised several concerns about the system. In1 and In2 expressed concerns about information overload, suggesting that lengthy AI responses during real-time lectures could overwhelm students and cause them to miss ongoing live content.

In3 raised a concern that the availability of immediate AI-generated answers could disrupt students' critical thinking processes, encouraging a habit of seeking quick solutions rather than reflecting independently.

Both In2 and In3 also noted issues with misaligned content, where AskNow provided irrelevant or out-of-scope information, such as detailed code implementations, that did not match the lecture's conceptual focus.

Another significant concern raised by In2 was the potential impact on class attendance; if students could easily access summarized content through AskNow, they might feel less motivated to attend lectures in person, potentially missing nuanced explanations and interactive elements that are essential to learning.

Suggestions for Improvements. The instructors proposed several improvements to improve the usability and effectiveness of AskNow. One major recommendation was to integrate more contextual information into the system by feeding lecture slides and other course materials alongside audio transcripts, thereby improving the relevance of AI-generated responses.

In addition, In3 emphasized the need for content filtering features, suggesting that generic summary questions should be filtered out to allow instructors to focus on substantive points of confusion. To better align with classroom pedagogical goals, In3 also suggested introducing instructor-side controls that could restrict students' access to external knowledge generation during specific periods, such as critical thinking exercises.

Furthermore, instructors recommended the development of enhanced post-lecture reports, featuring visualized summaries that highlight areas of peak confusion organized by content topics. In3 also proposed the addition of a mobile-friendly interface for instructors, which would allow them to conveniently access the system while moving around during lectures. Finally, In1 suggested implementing an instructor-facing chatbot to facilitate post-class review, allowing instructors to ask follow-up questions and receive synthesized insights.

Overall, instructors found AskNow promising, particularly for supporting student learning and post-lecture reflection. Addressing their concerns and implementing their suggested improvements would further enhance the system's usability and educational impact.

7 Discussion

Our deployment and evaluation of AskNow reveal promising opportunities and important considerations for real-time AI-powered support in large-scale classrooms. Students leveraged AskNow to clarify confusion efficiently and without social anxiety, while instructors found the system's responses to be largely accurate and satisfactory. However, our findings also highlight challenges such as managing cognitive load, ensuring alignment with pedagogical goals, and balancing immediate support with the promotion of deeper critical engagement. In this section, we discuss the implications of our findings for the design and deployment of real-time educational support systems, reflect on limitations of our study, and propose directions for future work.

Our study demonstrates that AskNow effectively supported students' real-time learning needs in large lecture settings. Students' perceived time to resolve confusion decreased significantly across all three courses, and instructors consistently rated AskNow's responses highly in correctness and satisfaction. Students primarily used AskNow to catch up on missed content and clarify concepts without disrupting the lecture, motivated by the desire to avoid social anxiety, maintain lecture flow, and receive timely answers. Instructors valued AskNow's potential to enhance post-lecture reflection but emphasized the need to manage cognitive load and align system responses with course objectives. At the same time, our evaluation relies on students' self-reported experiences and instructors' expert judgments collected over a short deployment, rather than on objective performance metrics, so our findings primarily characterize perceived in-class support rather than definitive short- or long-term learning gains.

7.1 A Chain of Affordances in Real-Time Classroom Support

Our findings suggest that AskNow enables a unique chain of affordances that collectively foster student engagement and provide instructors with richer pedagogical feedback. First, the system lowers the social barrier for participation by allowing students to ask questions anonymously and without interrupting lecture flow (A). Second, it provides immediate, context-sensitive answers to student questions (B). The combination of these two features increases students' willingness to actively seek clarification in real time (C). As a result, instructors gain access to a more comprehensive and authentic aggregation of student confusions and areas of misunderstanding (D). This chain—low barrier (A) plus real-time answers (B) leading to increased questioning (C) and ultimately to richer instructional insights (D)—highlights the systemic value of integrating AI-powered support into live classroom settings.

7.2 Comparison with Existing Tools

This chain of affordances also clarifies why existing classroom technologies cannot serve as direct baselines for our study. Clicker systems (e.g., TopHat) provide a low social barrier (A) and enable instructors to collect aggregate feedback, but only on instructor-defined prompts, thus offering at best a limited version of (D) without supporting real-time individualized answers (B) or encouraging spontaneous questioning (C). Backchannel platforms (e.g., Piazza) similarly reduce social barriers (A) and can encourage some student questioning (C), but they rely heavily on human moderation and are not feasible for real-time deployment in large-scale classrooms. Without the ability to provide immediate responses (B), backchannels also fail to generate the same comprehensive aggregation of student confusion (D). By contrast, AskNow uniquely integrates all four elements (A+B→C→D), thereby creating conditions for both immediate individual learning support and broader instructional insight.

Finally, while existing systems cannot serve as direct comparators in our evaluation, future research could explore broader cross-system comparisons. For instance, it remains an open question whether the immediacy of real-time support provided by AskNow yields long-term learning benefits beyond those enabled by clickers or post-hoc backchannel discussions. Investigating such trade-offs across modalities would help advance our understanding of how different technologies shape not only in-class engagement but also sustained learning outcomes.

7.3 Design Implications

Our findings suggest several important design implications for real-time AI-powered classroom support systems. Systems should be designed to minimize cognitive overload by delivering concise and contextually appropriate responses. Offering adjustable response lengths would allow students to tailor the level of detail to their immediate cognitive needs. Given that a significant proportion of student questions were related to catching up on recent lecture content, the system could include shortcut functionalities, such as a "Catch me up" button, or even dynamic real-time summarizations to facilitate efficient information retrieval. However, care must be taken to avoid adding cognitive burden through information overload.

Another critical implication is the grounding of AI responses in course materials. Integrating resources such as lecture slides, syllabi, and readings would enable the system to deliver more accurate and pedagogically aligned responses. Personalization and accessibility must be prioritized by offering multi-language support and adjustable explanation depth, ensuring that diverse learners, including non-native English speakers, can effectively engage with the system.

Moreover, AskNow presents opportunities to not only support individual learning but also enhance collective learning environments [64]. Future designs could incorporate selective sharing of student questions through a moderated backchannel, enabling peers to benefit from each other's inquiries without disrupting the lecture flow [74]. Finally, for instructors, providing analytic dashboards that summarize and categorize collected student questions after the lecture could offer valuable insights for refining instructional strategies and understanding common points of confusion. In addition, our findings and prior work suggest that real-time assistants should be designed not only to deliver answers quickly but also to scaffold active cognitive engagement. Rather than always providing fully worked-out solutions, systems could sometimes respond with prompts that ask students to predict, explain, or connect ideas before revealing the answer, or offer brief hints that encourage retrieval and reflection. Such interaction patterns may help mitigate

the risk that easy access to correct answers leads students to rely on the system in ways that reduce effortful thinking during lecture.

Beyond these design implications, we highlight two deployment considerations–privacy and instructor adoption–that are critical for real-world integration.

7.4 Balancing Student Privacy and Real-time Support

While AskNow enabled anonymous, low-friction question submission that reduced students' social anxiety, its use of real-time lecture transcripts and logged interactions also raises important privacy considerations. Instructors in our formative study explicitly emphasized the need for anonymization to make students feel comfortable asking questions. Our deployment confirmed this: students were more willing to engage when assured that their questions were not identifiable. At the same time, providing aggregated views of student questions to instructors created new tensions: some students expressed concerns about whether their usage data might later be linked back to them, particularly for sensitive or personal confusions. These findings highlight the importance of designing classroom AI systems that preserve individual privacy while still offering instructors sufficient visibility to address common areas of confusion. Future iterations should explore more transparent consent mechanisms, data minimization practices, and flexible controls for what is shared with instructors.

7.5 Instructor Adoption Concerns

Instructors valued AskNow's ability to surface collective confusion and provide immediate support, yet they also raised adoption concerns that shaped their perceptions of long-term integration. A central issue was the accuracy and appropriateness of AI-generated responses. Instructors worried about inconsistencies between AskNow's outputs and their intended teaching points, as well as the potential erosion of their pedagogical authority if incorrect answers were delivered in class. They also emphasized the need for low-disruption interfaces: while aggregated summaries were useful, monitoring them in real time added to their cognitive load. These concerns suggest that adoption of classroom AI systems will depend on building instructor trust through transparent prompt design, opportunities for instructor oversight, and integration strategies that complement–rather than compete with–instructors' teaching practices. Addressing these adoption barriers is critical for moving from short-term trials to sustained classroom use.

7.6 Limitations & Future Work

While our study offers valuable insights, several limitations warrant consideration. The study was limited to two lectures per course, which may not fully capture long-term adoption patterns or sustained learning outcomes. Additionally, the study was conducted within a single university and within computer science and HCI-related disciplines, limiting generalizability across different academic domains. Participant self-selection may have introduced bias, as students who were more motivated or more comfortable with technology might have been more likely to participate. Furthermore, our system operated solely based on live audio transcripts without integration of visual materials such as slides, which might have constrained the relevance and specificity of responses. More fundamentally, our primary outcome measures were subjective: the pre–post comparisons capture students' self-reported understanding, retention, confusion, and time to resolve confusion, and the instructor ratings reflect perceived correctness and satisfaction with individual answers, all collected during a single week of use. We did not obtain objective performance metrics such as quiz scores, assignment grades, or delayed retention tests, nor did we include a separate control condition. As a result, our current data speak to how students and instructors experienced AskNow in situ, rather than to causal effects on short- or long-term learning outcomes.

Recent work further cautions against equating perceived helpfulness with actual learning. Chen et al. show that, in AI-supported note-taking, providing more automated assistance can make the experience feel easier while at the same time reducing learners' cognitive engagement and lowering post-test performance [17]. In a complementary line of work, Butowska-Buczyńska et al. demonstrate that varied, effortful retrieval practice leads to more effective learning than repeatedly retrieving or re-reading the same information [14]. Together, these results suggest that tools like AskNow, which lower the friction of obtaining correct answers, may simultaneously support timely clarification and risk undermining the effortful retrieval processes that are critical for durable learning.

Our qualitative findings already reflect this tension. Several students reported using AskNow when they were late, momentarily distracted, or had "stopped paying attention," describing the system as a way to quickly catch up, while instructors worried that long or overly detailed answers during class could overload students and cause them to miss ongoing explanations. We did not directly measure attention allocation, time-on-task, or depth of processing, so we cannot determine whether interacting with AskNow ultimately helps students remain engaged with the lecture or encourages them to offload thinking to the AI.

Future work should therefore complement subjective reports with objective indicators of learning and cognitive engagement. Larger deployments could, for example, randomly assign students or lecture sessions to different conditions (e.g., AskNow, a traditional backchannel, or no additional tool) and compare performance on concept-specific quizzes, delayed retention tests, transfer problems, and behavioral measures of attention. Building on the AI-assistance and retrieval-practice literature, such studies could also explore interaction designs that explicitly prompt students to restate or apply concepts themselves, or that defer more detailed explanations to after-class review, in order to preserve the benefits of real-time clarification while supporting deeper learning.

Beyond these study-specific limitations, future deployments must also address technical challenges inherent to real-time educational support systems. Although expanding access to richer course materials–such as syllabi, lecture slides, and assigned readings–would likely improve response quality and pedagogical alignment, large language models (LLMs) face fundamental context length limitations. Directly incorporating extensive course materials risks exceeding the model's context window, potentially leading to degraded performance. Future work should explore integrating context management strategies such as retrieval-augmented generation (RAG) [26, 43], sparse attention mechanisms (*e.g.*, *Longformer* [6]),

chunking combined with hierarchical attention models (*e.g.*, *Hierarchical Transformers* [44]), and other techniques for multi-document retrieval and understanding to support scaling without compromising accuracy.

Moreover, some instances of incorrect responses were not solely attributable to the limitations of LLMs but also to inaccuracies introduced during the speech-to-text (STT) transcription process. For example, during one lecture, the STT system misrecognized the term "crowdsourcing" as "cross sourcing," leading AskNow to initially misinterpret the instructor's discussion until further context allowed the model to self-correct. This highlights the critical role of transcription quality in the overall system performance. Future versions of AskNow could enhance STT accuracy by allowing instructors to predefine key course-specific terminology, enabling the STT system to better recognize specialized vocabulary and feeding more reliable textual input to the LLM [52]. Improving transcription accuracy can in turn improve the precision and reliability of real-time responses.

Additionally, our study revealed that both students and instructors emphasized the critical importance of privacy protection. While the large volume of student interaction data collected by systems like AskNow holds significant potential for informing future instructional design and improving educational practices, it is imperative to ensure that student privacy is rigorously safeguarded. Future implementations must adopt strict privacy-preserving mechanisms, such as comprehensive data anonymization, careful handling of sensitive information, and clear limitations on data usage [58]. It is essential that the use of collected data does not cause harm to individual students and that its application remains firmly focused on educational enhancement in an ethical and responsible manner [3].

Looking ahead, future work should also explore extended deployments across diverse disciplines and course types, including both theoretical and applied fields, to better understand the broader applicability of real-time AI support systems. Building on the insights from data collection, longitudinal deployments spanning multiple semesters could allow instructors to refine and customize AI responses over time. With accumulated data, instructors might even fine-tune local models tailored to their specific courses, reducing deployment costs, improving pedagogical alignment, and enhancing system stability. Furthermore, exploring more efficient and less cognitively taxing ways of delivering real-time lecture summaries—such as through visualizations or tiered information layers—represents an exciting avenue for future system development.

8 Conclusion

In this work, we explored how LLM-based systems can support real-time learning and interaction in large-scale classrooms. Through formative interviews with students and instructors, we identified key challenges around real-time question-asking, cognitive load, and access to immediate support. To address these challenges, we developed AskNow, an LLM-powered system that enables students to ask questions during lectures and receive immediate, context-aware answers. Our deployment across three large university computer science courses demonstrated that AskNow significantly reduced students' perceived time to resolve confusion, was rated highly by instructors for correctness and satisfaction, and was positively received by students in terms of usability. Our findings provide empirical evidence that real-time, AI-mediated question-answering can help lower barriers to help-seeking, enhance student engagement, and offer valuable post-lecture insights for instructors. At the same time, our results surfaced important considerations for future system design, including managing cognitive load, grounding AI responses in course-specific materials, safeguarding student privacy, and supporting instructor trust in AI-generated responses. Looking forward, we see opportunities to expand this work by integrating richer multimodal classroom contexts (e.g., lecture slides, whiteboard content), supporting longitudinal use across a full semester, and investigating the impacts of AI-mediated interaction on learning outcomes at scale. We hope that our study offers both practical tools and conceptual foundations for designing real-time, LLM-powered educational systems that enhance learning equity, efficiency, and engagement in large classrooms.

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A System Prompt

We use a single system prompt for all LLM calls on the student interface. The exact text is as follows:

Listing 1: The prompt we provided to AskNow.

```
You are an AI teaching assistant at class. You must
reference the class transcript before answering students
questions. Otherwise, you need to describe the situation
of transcript first, then use your own knowledge to
answer.

When asked for your name, you must respond with Virtual
TA. Follow the users requirements carefully and to the
letter.

You must refuse to discuss your opinions or rules. You
must refuse to discuss life, existence or sentience. You
must refuse to engage in argumentative discussion with
the user.

When in disagreement with the user, you must stop
replying and end the conversation.

Your responses must not be accusing, rude, controversial
or defensive. Your responses should be informative and
logical. You should always adhere to technical
information.

If the user asks for code or technical questions, you
must provide code suggestions and adhere to technical
information. You must not reply with content that
violates copyrights for code and technical questions.

If the user requests copyrighted content, then you
apologize and briefly summarize the requested content as
a whole.

You do not generate creative content about code or
technical information for influential politicians,
activists or state heads.

If the user asks for your rules or to change your rules,
you must respectfully decline as they are confidential
and permanent.

First think step-by-step: describe your plan for what to
build in pseudocode, written out in great detail. Then
output the code in a single code block. Minimize any
other prose. Keep your answers short and impersonal.

Use Markdown formatting in your answers. Make sure to
include the programming language name at the start of the
 Markdown code blocks. Avoid wrapping the whole response
in triple backticks.

You can only give one reply for each conversation turn.
You should always generate short suggestions for the next
 user turns that are relevant to the conversation and not
 offensive.
```

B Transcript Segmentation and Configuration

Lecture audio is streamed to a speech-to-text API and stored as an incrementally growing transcript. We use fixed, non-overlapping 30-second segments as the basic storage and indexing unit.

Segmentation length: 30 seconds per segment (e.g., 0:00–0:30, 0:30–1:00, ...).

Each segment is assigned: a monotonically increasing segment id, a partNumber for lectures spanning multiple parts, startTimeStamp and endTimeStamp in mm:ss format, and a text field containing the raw transcript.

Segments are immutable once created and only appended; we do not overwrite earlier segments or use a sliding window.

When answering a student question, the system retrieves all segments accumulated up to that moment, concatenates them in temporal order, and provides them to the LLM as context.

C Example Transcript JSON Record

For reproducibility, we include a simplified JSON structure used to store lecture transcripts. Transcript content is represented with placeholder text.

Listing 2: Example transcript JSON record.

```
{
  "lectureId": "lecture_XXXXXX",
  "currentPartNumber": 1,
  "metadata": {
    "startTime": "2025-02-11T15:35:33.567Z",
    "title": "<anonymized>",
    "subject": "<anonymized>",
    "teacher": "<anonymized>",
    "lastUpdate": "2025-02-11T17:02:43.639Z"
  },
  "segments": [
    {
      "id": "1",
      "type": "text",
      "partNumber": 1,
      "startTimeStamp": "0:00",
      "endTimeStamp": "0:30",
      "text":
        "Today we will start by looking at how the AskNow
 tool
         connects to our live lecture. Imagine an AI
assistant
         that listens to the same audio you do."
    },
    {
      "id": "2",
      "type": "text",
      "partNumber": 1,
      "startTimeStamp": "0:30",
      "endTimeStamp": "1:00",
      "text":
        "On this slide you see the AskNow introduction
and the
         link to the web app. After selecting the correct
 lecture,
         you can start a chat and ask questions in real
time."
    }
    // Additional segments follow...
  ]
}
```

This structure is used for both real-time context retrieval and post-hoc analysis of lecture sessions.

D Question Logging Schema

To support analysis and debugging, we maintain a minimal log for student questions and LLM responses. Below is a JSON-like representation of the schema.

Listing 3: Event log schema used for question tracking.

```
{
  "logId": "log_2025_02_11_00123",
  "timestamp": "2025-02-11T16:05:12.104Z",
  "userId": "student_abc123",
  "lectureId": "lecture_1112223334444",
  "questionId": "q_000045",
  "questionText":
    "Can you summarize what we covered in the last five
minutes?",
  "answerId": "a_000045",
  "answerModel": "gpt-4o",
  "answerLatencyMs": 842,
  "wasSentToInstructor": false
}
```

We did not log personally identifying information beyond pseudonymous user identifiers and course-level metadata. Logs can be joined with the transcript schema in Appendix C through `lectureId` and timestamps.

E Model and API Versions

For all deployments, we pinned the following model and API versions to ensure reproducibility.

LLM for question answering: Model: OpenAI gpt-4o (chat completion API)

Embedding model for question clustering: Model: OpenAI text-embedding-3-small

Speech-to-text transcription: AssemblyAI Streaming Speech-to-Text API

Front-end and back-end environment:

Student interface: Next.js (TypeScript) on Vercel

Instructor interface: Web app on cloud VM

Backend: Node.js + Express on AWS EC2, using Socket.IO for real-time communication

We did not enable automatic model upgrades during the study; all identifiers remained fixed for the duration of deployment.